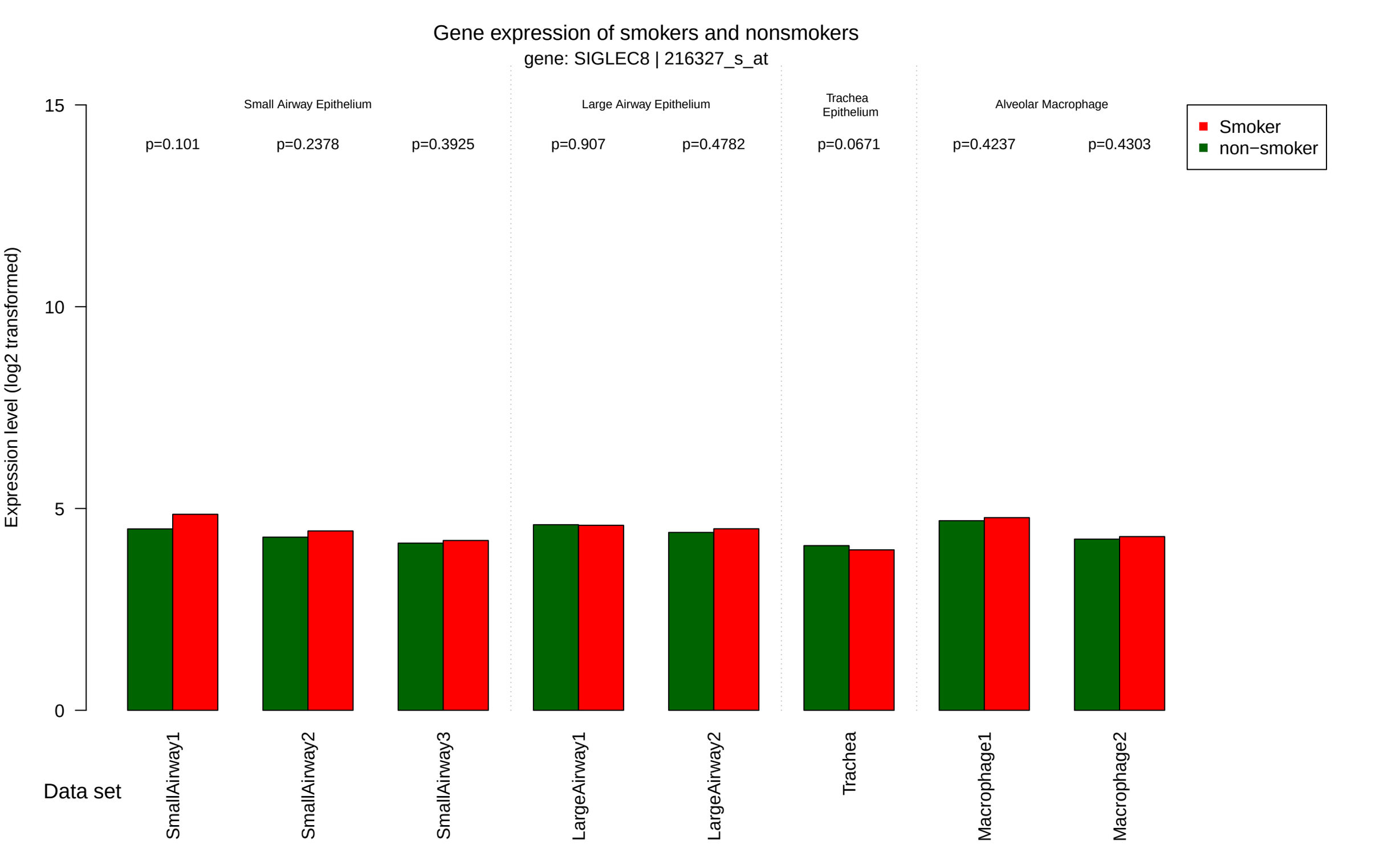
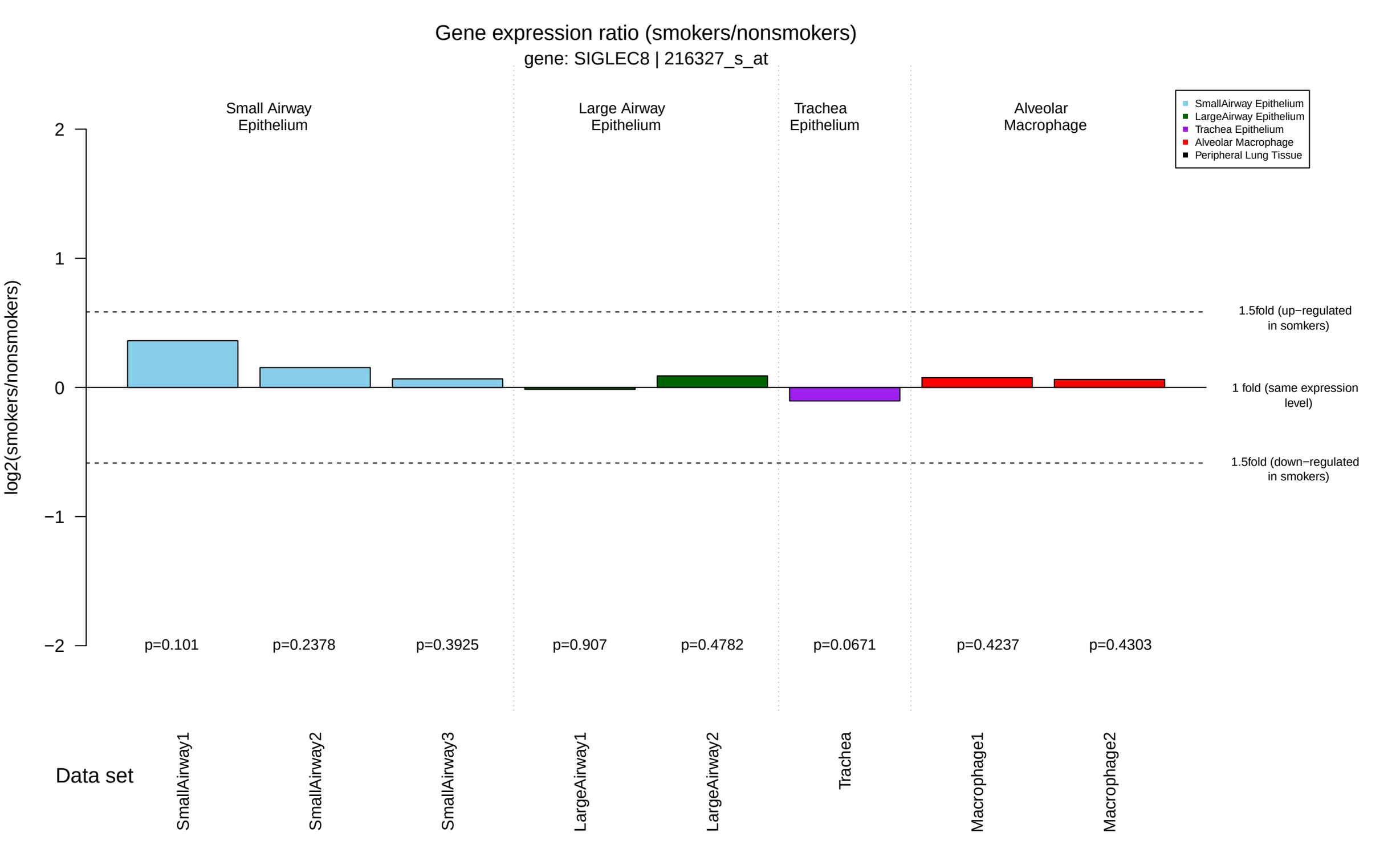
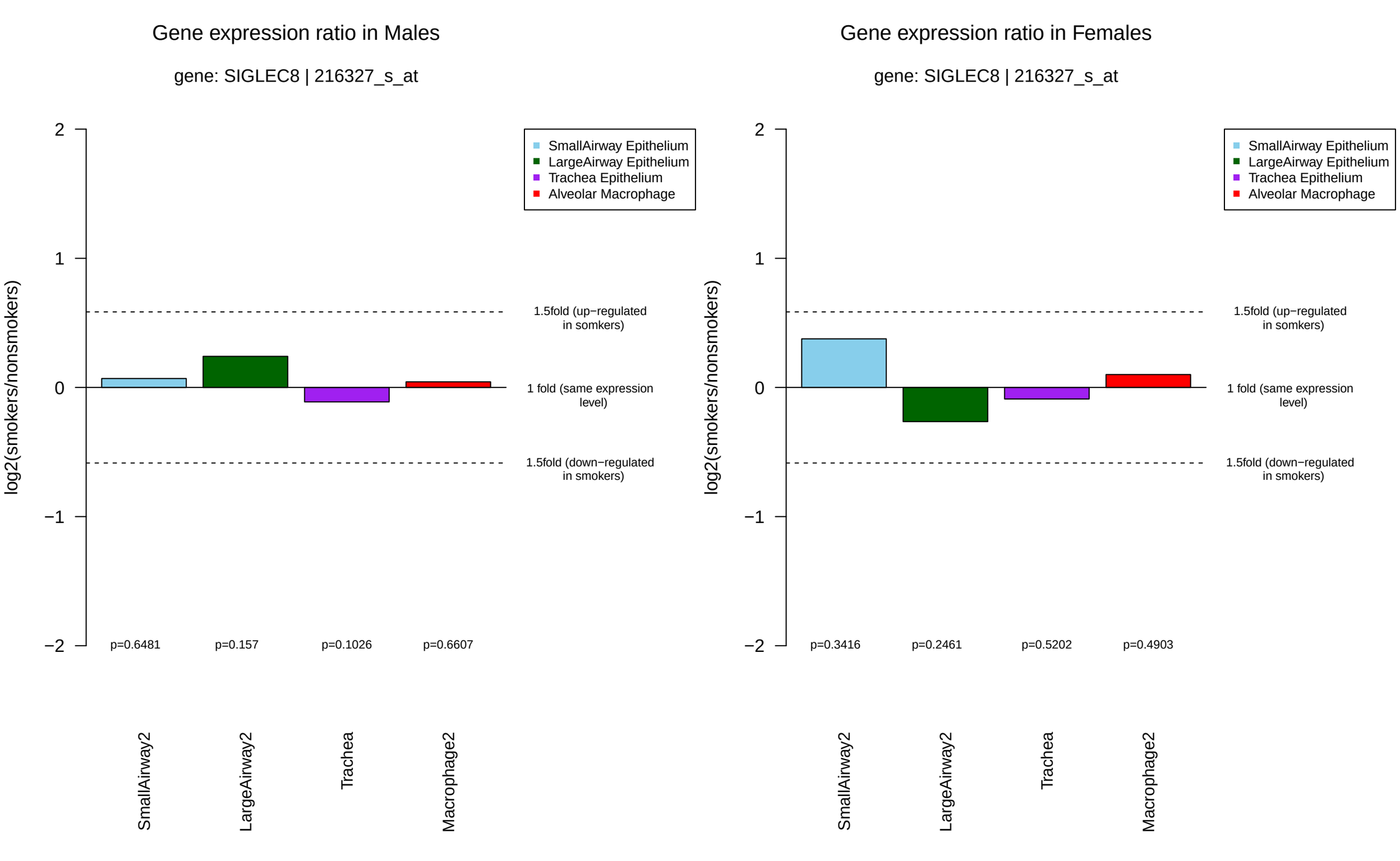
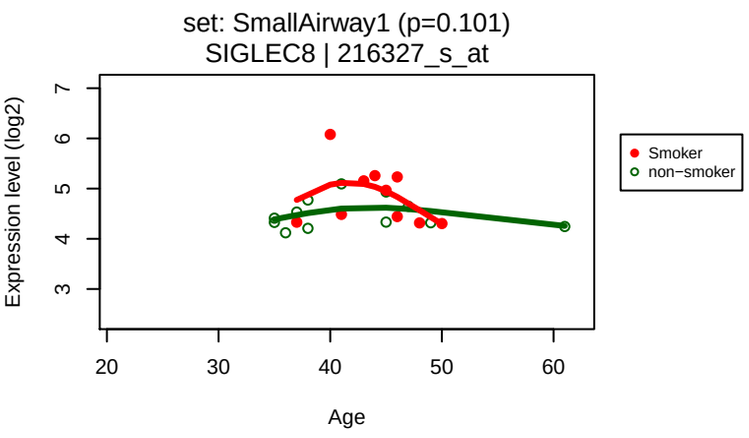
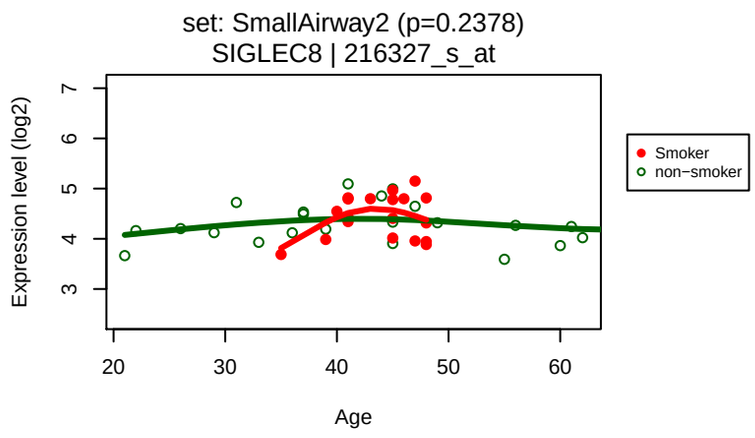
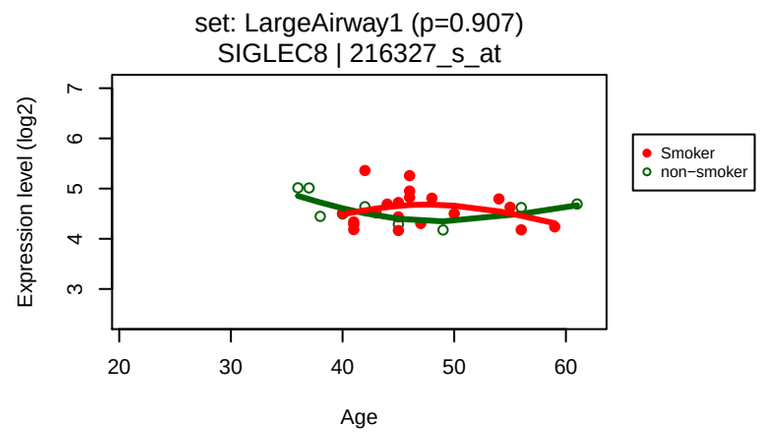

Gene expression of smokers and nonsmokers
gene: SIGLEC8 | 216327_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.101
p=0.2378
p=0.3925
p=0.907
p=0.4782
p=0.0671
p=0.4237
p=0.4303
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

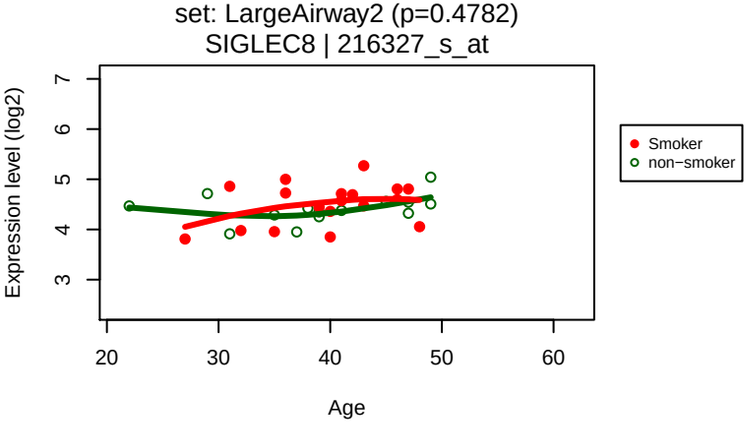

Gene expression ratio (smokers/nonsmokers)
gene: SIGLEC8 | 216327_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.101
p=0.2378
p=0.3925
p=0.907
p=0.4782
p=0.0671
p=0.4237
p=0.4303
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

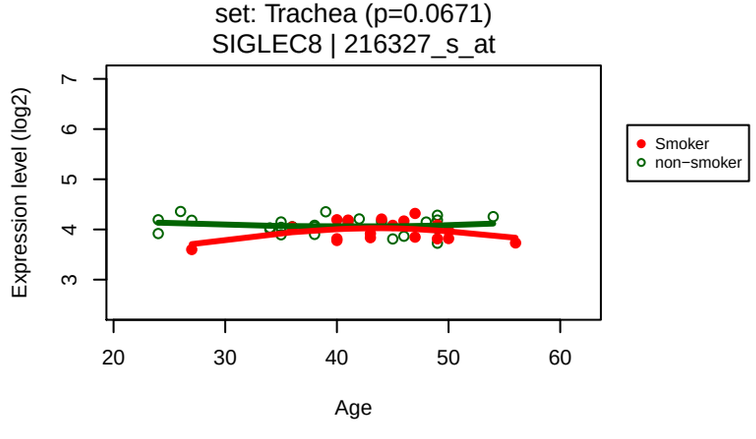

Gene expression ratio in Males
gene: SIGLEC8 | 216327_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6481
p=0.157
p=0.1026
p=0.6607
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIGLEC8 | 216327_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3416
p=0.2461
p=0.5202
p=0.4903
SmallAirway2
LargeAirway2
Trachea
Macrophage2

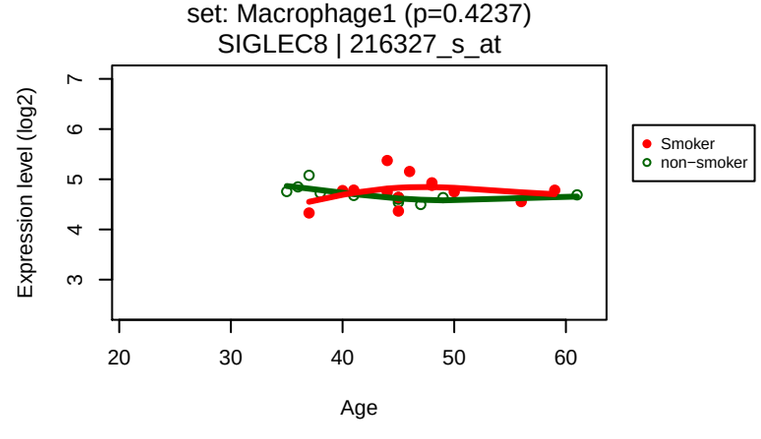

set: SmallAirway1 (p=0.101)
SIGLEC8 | 216327_s_at
Expression level (log2)
Age
Smoker
non−smoker

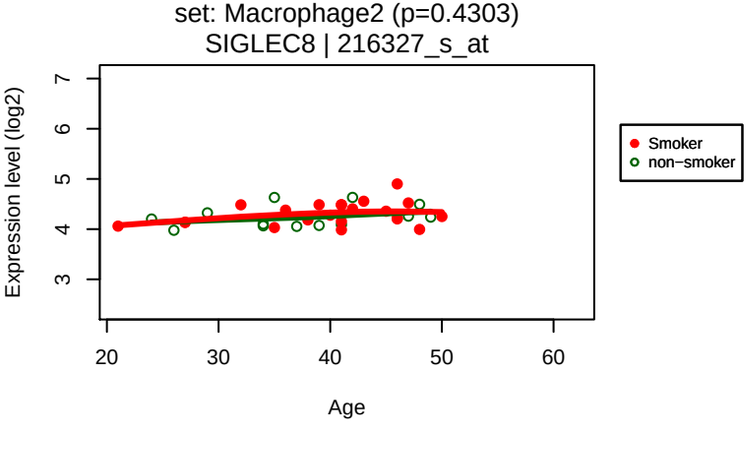

set: SmallAirway2 (p=0.2378)
SIGLEC8 | 216327_s_at
Expression level (log2)
Age
Smoker
non−smoker

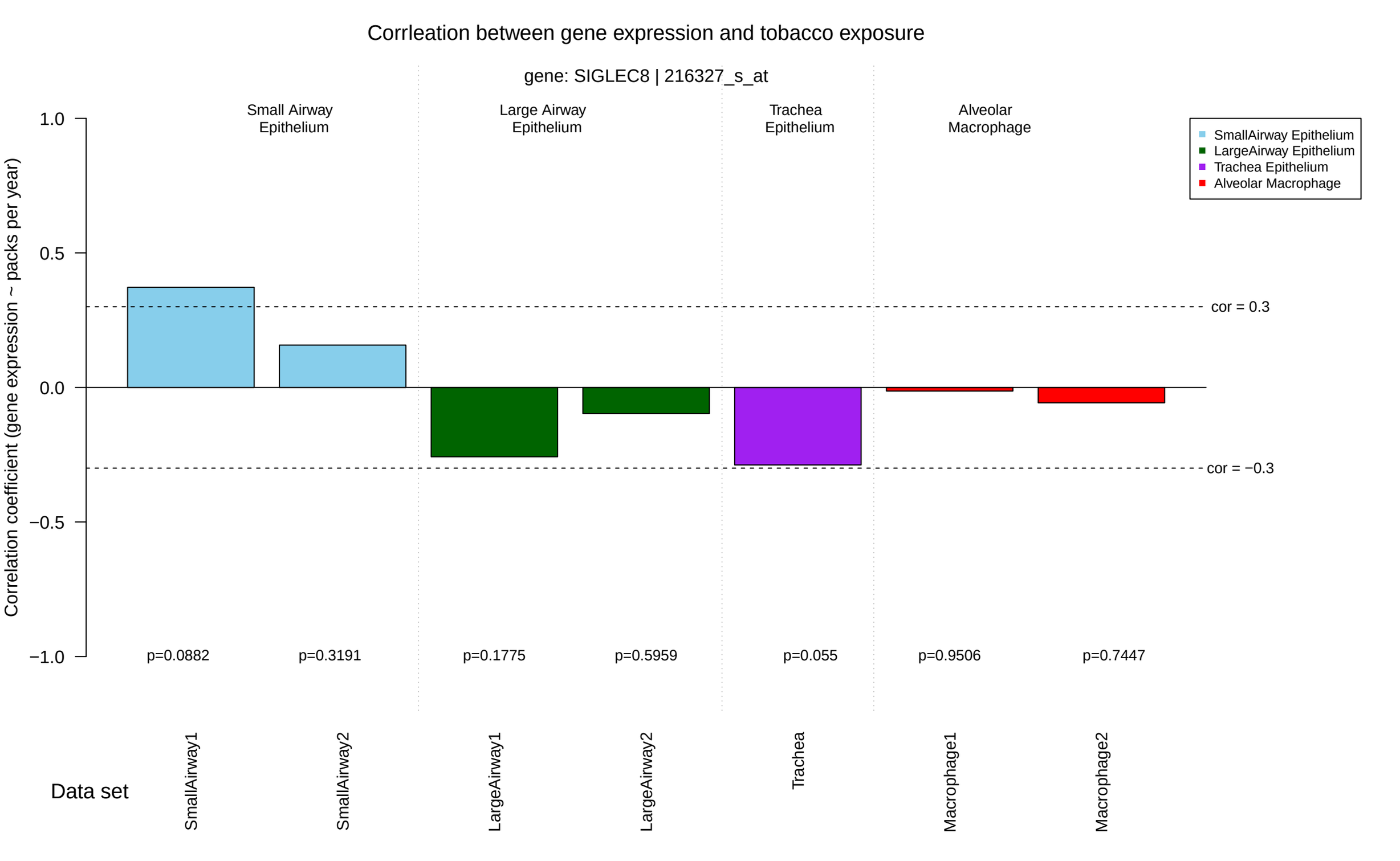

set: LargeAirway1 (p=0.907)
SIGLEC8 | 216327_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4782)
SIGLEC8 | 216327_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0671)
SIGLEC8 | 216327_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4237)
SIGLEC8 | 216327_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4303)
SIGLEC8 | 216327_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIGLEC8 | 216327_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0882
p=0.3191
p=0.1775
p=0.5959
p=0.055
p=0.9506
p=0.7447
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2